Exercise 1. Decide the convergence or divergence of the following series:

$$\text{a)}\sum_{n=1}^{\infty}\frac{3}{9n-1} \qquad \text{b)}\sum_{n=1}^{\infty}\frac{1}{2n-1} \qquad \text{c)}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} \qquad \text{d)}\sum_{n=1}^{\infty}\frac{1}{n(n+1)} \qquad \text{e)}\sum_{n=1}^{\infty}ne^{-n^2}$$

Proof. Start! We want to prove the convergence or divergence of the following:

a) By the Comparison Test,

$$\sum_{n=1}^{\infty}\frac{3}{9n-1}=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{3}{9n-1}\geq\lim_{N\to\infty}\sum_{n=1}^{N}\frac{3}{10n}=\sum_{n=1}^{\infty}\frac{3}{10n}=\infty;$$

otherwise,

$$\sum_{n=1}^{\infty}\frac{1}{n}=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n}=\lim_{N\to\infty}\frac{10}{3}\sum_{n=1}^{N}\frac{3}{10n}=\frac{10}{3}\sum_{n=1}^{\infty}\frac{3}{10n}<\infty.$$

b) By the Comparison Test,

$$\sum_{n=1}^{\infty}\frac{1}{2n-1}=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{2n-1}>\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{2n}=\sum_{n=1}^{\infty}\frac{1}{2n}=\infty;$$

otherwise,

$$\sum_{n=1}^{\infty}\frac{1}{n}=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n}=\lim_{N\to\infty}2\sum_{n=1}^{N}\frac{1}{2n}=2\sum_{n=1}^{\infty}\frac{1}{2n}<\infty.$$

c) By the Absolute Series Test,

$$\sum_{n=1}^{\infty}\frac{1}{n^2}<\infty\implies\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}<\infty.$$

d) By the Telescoping Series Test,

$$\begin{aligned}\sum_{n=1}^{\infty}\frac{1}{n(n+1)}&=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n(n+1)}\\&=\lim_{N\to\infty}\sum_{n=1}^{N}\left(\frac{1}{n}-\frac{1}{n+1}\right)\\&=\lim_{N\to\infty}\left(1-\frac{1}{N+1}\right)=1-\lim_{N\to\infty}\frac{1}{N+1}=1<\infty.\end{aligned}$$

e) By the Comparison Test,

$$\sum_{n=1}^{\infty}ne^{-n^2}=\lim_{N\to\infty}\sum_{n=1}^{N}\frac{n}{e^{n^2}}<\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n^2}=\sum_{n=1}^{\infty}n^{-2}<\infty;$$

in view of the fact,

$$\forall n\in\mathbb{N}:n<2^n\implies n^3<2^{3n}<e^{n^2}\implies\frac{n}{e^{n^2}}<\frac{1}{n^2}.$$

□

Exercise 2.

a) Prove if $\sum x_n$ converges, then $\sum(x_{2n} + x_{2n+1})$ also converges.

b) Find an explicit example where the converse does not hold.

Proof. Start!

a) We want to prove that

$$\sum_{n=1}^{\infty} x_n < \infty \implies \sum_{n=1}^{\infty}(x_{2n} + x_{2n+1}) < \infty.$$

If

$$\sum_{n=1}^{\infty} x_n = \lim_{N\to\infty} \sum_{n=1}^{N} x_n < \infty \implies \sum_{n=1}^{\infty} x_{n+1} = \lim_{N\to\infty} \sum_{n=1}^{N} x_{n+1} < \infty,$$

then

$$\sum_{n=1}^{\infty}(x_{2n} + x_{2n+1}) = \lim_{N\to\infty} \sum_{n=1}^{N}(x_{2n} + x_{2n+1}) = \lim_{N\to\infty} \sum_{n=1}^{2N} x_{n+1} < \infty.$$

b) We want to prove that

$$\sum_{n=1}^{\infty}(x_{2n} + x_{2n+1}) < \infty \implies \sum_{n=1}^{\infty} x_n \not< \infty.$$

If $x_n = (-1)^n$, then

$$\sum_{n=1}^{\infty}(x_{2n} + x_{2n+1}) = \lim_{N\to\infty} \sum_{n=1}^{N} 0 < \infty$$

and

$$\sum_{n=1}^{\infty} x_n = \lim_{N\to\infty} \sum_{n=1}^{N}(-1)^n \not< \infty.$$

□

Exercise 3. Prove the triangle inequality for series, that is if $\sum x_n$ converges absolutely, then

$$\left|\sum_{n=1}^{\infty} x_n\right| \leq \sum_{n=1}^{\infty} |x_n|.$$

Proof. Start! We want to prove inductively that

$$\forall n \in \mathbb{N} : \left|\sum_{j=1}^{n} x_j\right| \leq \sum_{j=1}^{n} |x_j|.$$

1) (Base Case) If $n = 1$, then

$$\left| \sum_{j=1}^{n} x_j \right| \leq |x_1| \leq \sum_{j=1}^{n} |x_n|.$$

2) (Inductive Step) Assume, for the purpose of induction,

$$\forall n = 1, 2, 3, ..., m : \left| \sum_{j=1}^{n} x_j \right| \leq \sum_{j=1}^{n} |x_j|.$$

By the Triangle Inequality,

$$\left| \sum_{j=1}^{n+1} x_j \right| = \left| \sum_{j=1}^{m} x_j + x_{m+1} \right| \leq \left| \sum_{j=1}^{m} x_j \right| + |x_{m+1}| \leq \sum_{j=1}^{m} |x_j| + |x_{m+1}| = \sum_{j=1}^{n+1} |x_j|.$$

We want to prove directly that

$$\sum_{j=1}^{\infty} |x_n| < \infty \implies \left| \sum_{j=1}^{\infty} x_j \right| \leq \sum_{j=1}^{\infty} |x_j|.$$

1) (Lemma Case) Assume, for the purpose of implication,

$$\sum_{j=1}^{\infty} x_j < \infty \implies \lim_{n \to \infty} \sum_{j=1}^{n} x_j < \infty.$$

By the Reverse Triangle Inequality, $\forall \epsilon > 0 \ \exists M \in \mathbb{N} \ \forall n \geq M$ such that

$$\left| \left| \sum_{j=1}^{n} x_j \right| - \left| \sum_{j=1}^{\infty} x_j \right| \right| \leq \left| \sum_{j=1}^{n} x_j - \sum_{j=1}^{\infty} x_j \right| < \epsilon;$$

hence,

$$\lim_{n \to \infty} \left| \sum_{j=1}^{n} x_j \right| = \left| \sum_{j=1}^{\infty} x_j \right| = \left| \lim_{n \to \infty} \sum_{j=1}^{n} x_j \right|.$$

2) (Theorem Step) If

$$\sum_{j=1}^{\infty} |x_j| < \infty \implies \sum_{j=1}^{\infty} x_j < \infty,$$

then

$$\left| \sum_{j=1}^{\infty} x_j \right| = \left| \lim_{n \to \infty} \sum_{j=1}^{n} x_j \right| = \lim_{n \to \infty} \left| \sum_{j=1}^{n} x_j \right| \leq \lim_{n \to \infty} \sum_{j=1}^{n} |x_j| = \sum_{j=1}^{\infty} |x_j|.$$

□

Exercise 4. Decide the convergence or divergence of the following series.

$$\text{a)} \sum_{n=1}^{\infty} \frac{1}{2^{2n+1}} \qquad \text{b)} \sum_{n=1}^{\infty} \frac{(-1)^n(n-1)}{n} \qquad \text{c)} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^{1/10}} \qquad \text{d)} \sum_{n=1}^{\infty} \frac{n^n}{(n+1)^{2n}}$$

Proof. Start! We want to prove the convergence or divergence of the following:

a) By the Geometric Series Test,

$$\sum_{n=1}^{\infty} \frac{1}{2^{2n+1}} = \lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{2^{2n+1}} = \frac{1}{2} \lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{4^n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{4^n} = \frac{1}{6} < \infty.$$

b) By the Divergence Test,

$$\begin{aligned} \lim_{n\to\infty} \frac{(-1)^n(n-1)}{n} &= \lim_{n\to\infty} (-1)^n \left(1 - \frac{1}{n}\right) \\ &= \lim_{n\to\infty} (-1)^n \neq 0 \implies \sum_{n=1}^{\infty} \frac{(-1)^n(n-1)}{n} \not< \infty. \end{aligned}$$

c) By the Alternating Series Test,

$$\forall n \in \mathbb{N} : \frac{1}{n^{1/10}} > \frac{1}{(n+1)^{1/10}} \ \wedge \ \lim_{n\to\infty} \frac{1}{n^{1/10}} = 0 \implies \sum_{n=1}^{\infty} \frac{(-1)^n}{n^{1/10}} < \infty.$$

d) By the Root Test,

$$\lim_{n\to\infty} \sqrt[n]{\frac{n^n}{(n+1)^{2n}}} = \lim_{n\to\infty} \frac{n}{n^2+2n+1} = \lim_{n\to\infty} \frac{1}{n+2} = 0 < 1 \implies \sum_{n=1}^{\infty} \frac{n^n}{(n+1)^{2n}} < \infty.$$

□

Exercise 5.

a) Suppose $\sum x_n$ converges and $x_n \geq 0$ for all $n \in \mathbb{N}$. Prove that $\sum x_n^2$ converges.

b) Find a series such that $\sum x_n$ converges but $\sum x_n^2$ diverges.

Proof. Start!

a) Let $\sum x_n$ converge and $x_n \geq 0$ for all $n \in \mathbb{N}$. Then $\sum x_n$ converges absolutely and

$$\lim_{n\to\infty} |x_n| = 0 \implies \exists M_0 \in \mathbb{N} \ \forall n \geq M_0 : |x_n| < 1 \implies |x_n|^2 < |x_n|.$$

By the Comparison Test,

$$\sum x_n^2 = \sum |x_n|^2 < \sum |x_n| = \sum x_n < \infty.$$

Thus, $\sum x_n^2$ converges (absolutely), as needed.

b) Let

$$\forall n \in \mathbb{N} : x_n = \frac{(-1)^n}{\sqrt{n}} \in \mathbb{R} \implies x_n^2 = \frac{1}{n} \in \mathbb{R}.$$

By the Alternating Series Test,

$$\forall n \in \mathbb{N} : |x_n| = \frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n+1}} \ \wedge \ \lim_{n\to\infty} |x_n| = \lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0 \implies \sum x_n < \infty.$$

By the Harmonic Series Test,

$$\sum x_n^2 = \sum \frac{1}{n} = \infty.$$

Thus, $\sum x_n$ converges and $\sum x_n^2$ diverges, as needed.

□